Pyramakerz®

Unit 3

Lesson 1

Beyond Imagination.

Beep & Go! Guiding the Way.

Time Frame: [90 min]

Lesson Objectives:

By the end of this lesson students will be able to:

- Explain how sound can be used to help visually impaired people navigate safely.
- Design a simple sound-guided path using buzzers and push buttons.
- Use addition and subtraction to calculate distances between buzzers on the sound-guided path.
- Describe how community helpers, such as city planners and engineers, use sound to assist people with visual impairments.

STEAM SCOUTS

Challenge questions of the lesson

1. If you want the buzzer to make sound only when someone is holding the button, is a push button or an on/off switch better? Why?
2. You walk 5 steps, then hear a buzz. Then you walk 3 more steps and hear another buzz. How many steps did you walk in total? If you want to make the next buzz 2 steps closer, how many steps would that be?

SEL Relation

- Social Awareness (Empathy and Perspective-Taking):
  - By designing a sound-guided path, students are directly engaging in perspective-taking, imagining the challenges faced by individuals with visual impairments.
- Responsible Decision-Making:
  - Students engage in ethical consideration by designing solutions that improve the safety and independence of others.
  - They make responsible choices when planning their path, considering user needs and safety during the circuit building process.
- Problem-solving skills:
  - are honed as they troubleshoot their circuits and adjust their path plans, demonstrating persistence and adaptability.
- Relationship Skills (Collaboration):
  - Working in pairs or small groups during planning and building phases (e.g., "Explorer" and "Guide" roles) strengthens collaboration and communication skills.

Engage

Teacher Role:

- **Empathy Builder:** Guide students to understand the challenges faced by children with visual impairments when navigating a playground.
- **Problem Introducer:** Clearly present the core problem: "How can we help children who can't see move around safely at the playground?"
- **Idea Facilitator:** Encourage and affirm students' initial brainstorming ideas, especially those related to using sound for guidance.
- **Questioner:** Pose open-ended questions to deepen their thinking about practical solutions and the tools needed.
- **Connector:** Link their ideas (like using a buzzer) to the concept of circuits and buttons, setting the stage for future exploration.

Expected From Students:

- **Active Listening:** Students will listen attentively to the dialogue and identify the problem being discussed.
- **Empathy:** Students will consider the perspective of children who cannot see and understand their needs on a playground.
- **Idea Generation:** Students will brainstorm initial ideas for how sound could be used to guide someone.
- **Participation in Discussion:** Students will share their thoughts, suggest solutions, and respond to teacher and peer questions.
- **Problem-Solving (Initial):** Students will begin to think about how a device could make a sound when a button is pressed.

Parts of the books included.


Pyramakerz®

Ms. Sara: “Imagine you're at the playground, but you can't see where to go. How can we help children who can’t see move around safely?”

Adam: “Maybe we can use sounds to guide them!”

Laila: “Like a special beep that tells them where to go!”

Khaled: “What if they press a button, and it makes a sound to show them the way?”

Ms. Sara: “That’s a great idea! What can we use to make a sound when someone pushes the button?”

Adam: “A buzzer! It makes a loud beep!”

Laila: “We can make a path with buttons and buzzers so they can follow the sounds!”

Ms. Sara: “Great thinking! Let’s design a smart playground with sound-guided paths!”

Story

- Set the Scene with Imagination:
  - o Teacher says:

"Close your eyes for a moment and imagine you're at the playground. It's busy and full of laughter—but you can't see where to go. How would you feel? How would you know where to walk or play safely?"

- Engage with the Story:

Read the story aloud with excitement and varied voice tones:

  - o Use a thoughtful tone for Miss Sarah,
  - o A curious or problem-solving tone for Adam, Laila, and Khaled,
  - o Add expression when mentioning sounds like "beep" or "buzzer."
- Prompt a Thinking Question:
  - o Ask: "What could we design to help children who can't see find their way around the playground?"
  - o Let students share quick ideas to activate prior knowledge.

Possible Strategies for Implementation:

- Role-Play Simulation:
  - o Let students close their eyes and try following a path with guided sound (teacher claps or uses a sound cue).
  - o Debrief: "How did that feel? Was the sound helpful?"
- Think-Pair-Share:
  - o Pose a guiding question like: "What would help kids who can't see play safely?"
  - o Students think quietly, then pair up and discuss, followed by class sharing.
- Anchor Chart Creation:
  - o Start a class chart titled: "Smart Playground Ideas" and list student ideas like buzzers, buttons, paths with sounds, or textured flooring.

Let's Think!

Objective: Students will explore how sound-based tools and technologies help people who are blind or visually impaired navigate public spaces safely. This activity builds awareness, empathy, and understanding of assistive technologies used by community helpers.

Possible Correct Answers:

☑ Buzzers on sidewalks tell us where to go.

- ☑ Traffic lights make sounds to say when to stop or go.
- ☑ Smartphones can talk to us and guide our way.
- ☑ Beacons in buildings help us find rooms and exits.
- ☑ Headsets give directions using sounds.

Not Correct (Intended distractors):

- ☐ Flashing lights tell us where to go. (Visual aid, not sound)
- ☐ Talking walls give directions. (Not common or realistic)
- ☐ Loud music plays everywhere to help us hear better. (Untrue and confusing)
- ☐ Colorful signs show the way. (Visual, not audio)

Possible Strategies for Implementation:

1. **Think-Pair-Share**
   - Have students think quietly about how sound helps people move safely.
   - Pair them to discuss what they've seen in real life (e.g., crosswalk beeping).
   - Share with the class and explain why each option helps or doesn't.
2. **Multisensory Learning**
   - Use audio examples: Play a sound clip of a crosswalk beep, talking GPS, or traffic signal.
   - Ask students: "What do you think this sound is helping someone do?"
3. **Role Play**
   - Simulate a simple navigation activity where one student is blindfolded and the other uses only sound cues (claps, buzzers, or voice commands) to guide them safely across the classroom.
4. **Anchor Chart Creation**
   - Create a chart titled "How Sound Helps Us" and add examples students give.
   - Place a checkmark next to tools that support accessibility for people who can't see.
5. **Community Helper Connection**
   - Connect answers to roles: e.g., city planners, engineers, and accessibility designers use sound cues in public spaces.

Explorer's Toolkit

Teacher Role:

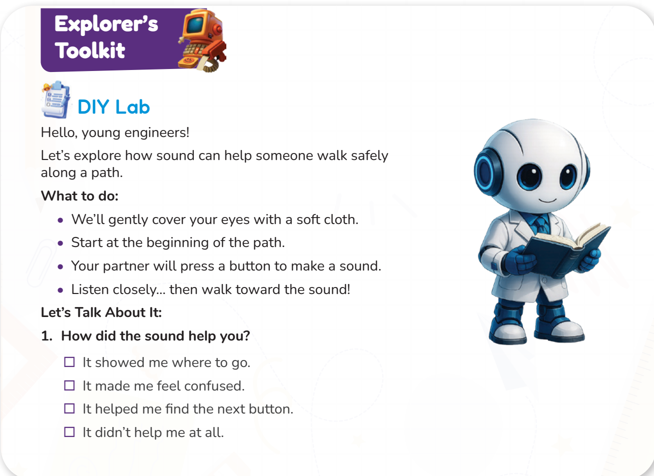
- **Introducer:** Clearly explain the purpose of the activity: to experience how sound can help someone navigate a path, especially when they can't see.

- **Safety Monitor:** Emphasize and ensure strict adherence to safety guidelines, particularly regarding covering eyes and guiding partners gently. Supervise closely to prevent any accidents or discomfort.
- **Instruction Giver:** Provide clear, step-by-step instructions for setting up the path, covering eyes, making sounds, and walking towards the sound.
- **Observer & Recorder:** Observe student interactions and how they use sound cues. Note any insights or challenges students express during the activity.
- **Discussion Facilitator:** Lead the "Let's Talk About It" discussion, prompting students to reflect on their experience and answer the questions.
- **Empathy Builder:** Encourage students to articulate how the experience of navigating by sound felt and why it's important for creating inclusive environments.

Expected From Students:

- **Active Listening:** Students will listen carefully to instructions and to the sounds made by their partners.
- **Safe Participation:** Students will follow safety rules, especially when their eyes are covered or when guiding a partner.
- **Empathy Engagement:** Students will actively engage in the simulation, putting themselves in the shoes of someone who cannot see.
- **Sound Following:** Students will attempt to navigate the path by listening and moving towards the sound.
- **Reflection & Articulation:** Students will reflect on their experience and answer the provided questions, articulating how sound helped or didn't help.

Parts of the books included.


Explorer's Toolkit

DIY Lab

Hello, young engineers!

Let's explore how sound can help someone walk safely along a path.

What to do:

- We'll gently cover your eyes with a soft cloth.
- Start at the beginning of the path.
- Your partner will press a button to make a sound.
- Listen closely... then walk toward the sound!

Let's Talk About It:

1. **How did the sound help you?**
   - ☐ It showed me where to go.
   - ☐ It made me feel confused.
   - ☐ It helped me find the next button.
   - ☐ It didn't help me at all.

DIY Lab

Objective: Students will explore how sound can assist people with visual impairments in navigating safely by participating in a guided activity. They will reflect on how sound provides directional cues, fostering empathy and design thinking for accessible environments.

Materials Needed:

- A soft cloth or blindfold to gently cover students' eyes
- A simple buzzer or button that makes sound
- Pre-set path on the classroom floor (e.g., tape lines, cones, or chairs as markers)
- Partner worksheet for responses
- Space clear of obstacles for safe walking

Activity Instructions:

1. **Set Up the Path:**
    - Create a simple walking path using tape, rope, or classroom furniture.
    - Place a buzzer or sound-emitting device at the end or along checkpoints.
2. **Assign Partners:**
    - One student wears a blindfold.
    - Their partner stays at the end of the path with the sound button.
3. **Start the Walk:**
    - The blindfolded student listens carefully and tries to walk toward the sound.
    - The partner presses the buzzer once every few seconds as a cue.
4. **Switch Roles:**
    - Allow both students to experience walking and guiding.
5. **Reflection Questions (Let's Talk About It):**

How did the sound help you?

What if there were no sounds?

Possible Strategies for Implementation:

- **Model First:** Show the class how to walk carefully and safely before blindfolding.
- **Safety Check:** Clear the path completely and have adults assist if needed.
- **Build Empathy:** Encourage students to describe how they felt during the walk.
- **Anchor Chart Reflection:** After the activity, make a class chart on "How Sound Helps People See."
- **Extend with Design Thinking:** Ask students how they might design sound-guided features for a playground.

Answers of the reflection questions:

1. **How did the sound help you?**

Answer:

☑ It showed me where to go.

☐ It made me feel confused.

- ☐ It helped me find the next button.
- ☐ It didn't help me at all

2. **What if there were no sounds?**

Answer:

- ☑ I wouldn't know which way to go.
- ☐ I could still find the path easily.
- ☑ I might get lost.
- ☑ I might walk the wrong way.

Explain

Teacher Role:

- **Contextualizer:** Explain that even though our main project is about sound and circuits, math helps us make our designs even better. Introduce the idea that adding and subtracting can help us plan paths and count steps.
- **Video Facilitator:** Introduce the "Adding and Subtracting" video as a tool to refresh or introduce these math skills, explaining that they will be useful for their "sound road" project.
- **Concept Clarifier:** After the video, explicitly review the definitions of addition and subtraction as "putting numbers together" and "taking away".
- **Application Guide:** Guide a discussion on how these math skills can be applied to designing a sound-guided path, specifically mentioning counting steps and measuring paths.
- **Encourager:** Emphasize how integrating math makes their "safe sound road even smarter!".

Expected From Students:

- **Active Listening:** Students will listen attentively to the teacher's explanation of how math connects to the project.
- **Video Engagement:** Students will watch the "Adding and Subtracting" video with a focus on understanding the concepts.
- **Math Concept Recall/Learning:** Students will grasp or review the basic definitions of addition and subtraction.
- **Application Thinking:** Students will think about how they can use addition and subtraction to plan and measure their sound-guided paths.
- **Discussion Participation:** Students will share their ideas on how math can make their projects better.

Parts of the books included.

Screen Lessons

Let's watch this video to learn more about adding and subtracting. It will help us make our road even better.

https://www.youtube.com/watch?v=NHI0ePgwlgU

Figure 32: A girl and a boy learning about adding and subtracting with fun math symbols.

Screen Lessons

Objective: Students will strengthen their understanding of addition and subtraction by observing how these operations are used in real-life situations presented in the video. This activity helps build number sense, supports mental math strategies, and encourages students to apply math thinking to solve everyday problems.

Video link: https://www.youtube.com/watch?v=NHI0ePgwlgU

Possible Strategies for implementation:

- Pre-Watching: Set a purpose.
  - Strategy: "Math Mission": Before playing the video, tell students: "Today, we're going on a 'math mission' to make our sound-guided playground paths even better! The video will help us remember or learn about adding and subtracting, which are super important for counting and measuring." Ask: "What do you already know about adding? What about taking away?"
  - Benefit: Frames the video within the context of their STEAM project and activates prior math knowledge.
- During Watching: Active engagement.
  - Strategy: "Count Along/Trace": Encourage students to count along or physically mimic the "adding" (putting together) and "subtracting" (taking away) actions shown in the video. If the video uses visuals, they can trace numbers or objects in the air.
  - Benefit: Promotes active engagement with the mathematical concepts presented in the video.
- Post-Watching: Discussion and reflection.
  - Strategy 1: "Recap the Math": Ask questions like: "What did the video teach us about 'adding'? What about 'subtracting'?" Use the definitions from the "Wrap-Up Time!" image to confirm their understanding: "Remember, addition means putting numbers together, and subtraction means taking away."
  - Benefit: Consolidates the mathematical learning from the video.
  - Strategy 2: "Math for Our Road": Lead a discussion connecting the math concepts directly to the "sound road" project. Ask: "How can we use addition to count the steps on our path? How can we use subtraction if we want to make our path

shorter or take away some sound buttons?" Emphasize: "We use these math skills to help us count steps, measure paths, and make our safe sound road even smarter!"

- o Benefit: Explicitly links the math concepts to the practical application in their ongoing STEAM project, reinforcing the interdisciplinary nature of STEAM.

Elaborate

Teacher Role:

- **Facilitator:** Introduce the "Monster Mansion Match Math Match" game as a playful and meaningful way to reinforce addition and subtraction concepts.
- **Guide:** Support students as they transition from the digital game to the real-world "Let's Plan Our Sound Path!" activity. Help them understand how math applies to planning inclusive playground features.
- **Prompt and Support:** Offer helpful questions and encouragement as students count steps, add and subtract distances, and problem-solve their layout.
- **Observer & Assessor:** Watch for how students engage with the game and apply math skills in the path planning activity. Note accuracy in calculations and thoughtful placement of sound features.
- **Motivator:** Celebrate students' use of math in real-world design and highlight the importance of inclusive thinking.

Expected From Students:

- Engage actively with the math game to strengthen their addition and subtraction skills.
- Apply what they learned by planning a sound path that includes logical placement of sound buttons.
- Use counting and calculations to decide the best distances between buttons and ensure the path is helpful for a visually impaired child.
- Record measurements and reasoning in a clear way.
- Share their plan and explain how math helped them make it fair and functional for all users.

Parts of the books included


Apprentice
Let's have fun with addition and subtraction by playing this exciting game!
https://www.abcya.com/games/math_match

MONSTER MANSION MATCH
MATH MATCH

Figure 33: Interface of a math game called "Monster Mansion Match."

Builder
Let's Plan Our Sound Path!
- Choose where you want to build your path.
- Think about the best places to put each sound button.

Figure 34: A colorful playground with a path for planning sound button placements.

- Walk from one button spot to the next — count how many steps you take.

Apprentice

Objective: Students will practice and reinforce their addition and subtraction facts (within 20) by playing a digital matching game. This activity strengthens math fluency, memory, and focus—skills needed when planning distances, counts, or quantities in playground design.

Activity link: https://www.abcya.com/games/math_match

Possible Strategies for Implementation:

1. **Set a Purpose Before Playing**
   - Tell students: “This game will help you get faster at adding and subtracting. You’ll need these skills later when we design and build parts of the playground like paths and ramps!”
   - You can write on the board: “Today I’m practicing math to become a better builder!”
2. **Use as a Warm-Up or Brain Break**
   - Let students play in pairs or independently for 5–10 minutes to boost engagement before moving into a related math or design task.
3. **Timed Challenge**
   - Challenge students to complete a level in a set time, then try to beat their score or time. This builds fluency and focus.
4. **Math Match Journal Prompt**
   - After playing, ask students to answer: “What kinds of math facts did you notice the most? Which ones were hardest to match?” This builds metacognition.
5. **Connect to Real-World Task**
   - After playing, transition to a design activity by saying: “Now that you’ve practiced your math brain, let’s use it to plan part of our playground! You’ll need addition and subtraction to figure out spaces and patterns.”

Builder

Objective: Students will apply addition and subtraction skills to plan and measure a sound-guided path, counting steps between sound buttons to ensure accessibility and logical layout. This builds connections between math and inclusive design while developing spatial reasoning and planning skills.

Possible Answers:

- “It took 6 steps between button 1 and 2.”
- “There are 18 steps total on my sound path.”
- “There are 2 more steps between button 2 and 3 than button 1 and 2.”

Activity Instructions

1. **Introduce the Goal**

Say: “Today we’ll build a path using sound buttons to help kids who can’t see find their way. We’ll use math to plan how far apart each sound should go.”

2. **Explain the Materials Needed**

- Sound button placeholders
- Playground cardboard base
- Marker or tape
- Rulers or measuring tape (optional)

3. **Guide Step-by-Step**

- Have students choose a starting point for their path.
- Ask them to place buttons at logical intervals along the path.
- Students will walk a toy and count how many steps it takes between each button.
- Record each segment’s steps on the dotted lines provided.
- Students then add all steps to calculate the total path length.
- Subtract to compare distances (e.g., “How many more steps between button 2 and 3 than between 1 and 2?”)

4. **Prompt Discussion**

Ask: “Where would someone need help the most?” or “Is there a better place to put the next sound?”

Possible Strategies for Implementation

1. **Peer Collaboration**

- Let students work in pairs — one walks, one counts and records.

2. **Use of Visual and Tactile Cues**

- Use masking tape or chalk to mark button spots physically.

3. **Anchor Chart Support**

- Display visuals with number lines, symbols (+ / –), and step vocabulary (e.g., total, more than, fewer than).

4. **Math Vocabulary Focus**

- Remind students to use math words: add, subtract, total, difference, count, steps.

Evaluate

Teacher Role:

- **Builder Facilitator:** Guide students through the process of physically assembling the button and buzzer circuit, ensuring they follow the diagram accurately.
- **Safety Supervisor:** Emphasize and monitor safe handling of all electronic components,

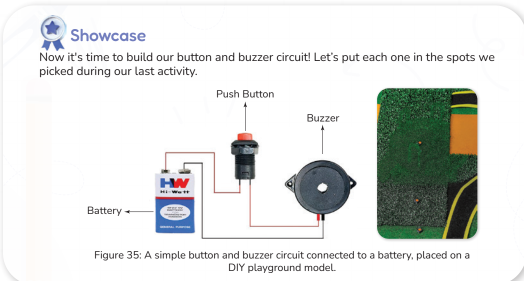
especially the battery. Ensure proper connections are made to avoid short circuits or component damage.

- **Problem-Solving Coach:** Provide direct support and troubleshooting assistance as students encounter challenges in connecting wires or getting their circuit to work. Encourage them to re-examine the diagram and their connections.
- **Encourager of Precision:** Stress the importance of careful, neat connections for the circuit to function correctly.
- **Evaluator:** Use the provided rubric to assess students' circuit building skills, functionality, and their ability to integrate the circuit into their planned path.
- **Discussion Leader:** Prompt students to reflect on the process of building, the challenges they faced, and how the circuit fulfills the purpose of sound guidance.

Expected From Students:

- **Circuit Assembly:** Students will physically connect the battery, button, and buzzer according to the provided circuit diagram.
- **Component Integration:** Students will place their built circuit (button and buzzer) into the pre-selected spots from their "sound path" planning activity.
- **Troubleshooting (with guidance):** Students will attempt to identify and correct simple errors in their circuit connections if it does not work initially.
- **Testing and Demonstration:** Students will test their circuit to ensure the buzzer makes a sound when the button is pressed.
- **Safe Handling:** Students will handle all components safely and responsibly.
- **Demonstration of Understanding:** Students will be able to explain how their button and buzzer work together to create a sound cue.

Parts of the books included.


Showcase

Now it's time to build our button and buzzer circuit! Let's put each one in the spots we picked during our last activity.

Figure 35: A simple button and buzzer circuit connected to a battery, placed on a DIY playground model.

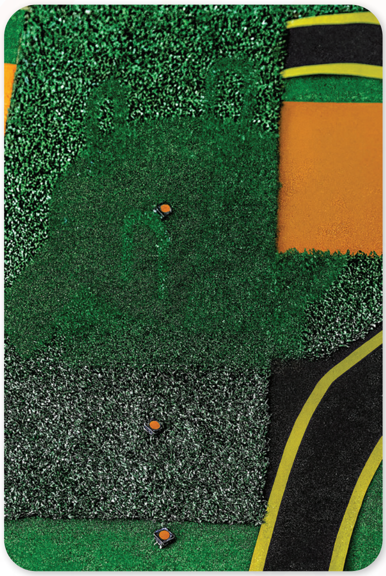
Showcase

Objective: Students will apply their understanding of sound-based accessibility by building a functional button-and-buzzer circuit and installing it along their planned sound path. This activity reinforces knowledge of circuits, promotes hands-on problem-solving, and connects inclusive design to real-world engineering

Materials Needed

- battery
- Buzzer

- Push button switch
- Wires with clips or jumper wires
- Cardboard or foam board model of the sound path (from previous activity)
- Tape or glue (for securing components)
- Markers or labels (optional, for button zones)



Activity Instructions

1. **Connect to Previous Learning**
   - Say: "Remember the spots you chose for the sound buttons in your path? Now we're going to bring them to life with a buzzer circuit!"
2. **Guide the Circuit Build**
   - Demonstrate how to connect the battery, switch, and buzzer in a simple circuit.
   - Emphasize safety (low-voltage batteries only).
   - Let students test their circuit before attaching it to the model.
3. **Install on the Path Model**
   - Instruct students to place each buzzer circuit where they had planned (on or near the cardboard/foam board path).
   - Use tape or glue to secure the components.
   - Let students take turns pressing the button and listening to how the circuit works.
4. **Encourage Reflection & Testing**

Final Prototype:

Prototype video link: https://drive.google.com/file/d/1Imt0Qx5c2qEtDYsipE4JU28-i4OMMtDF/view

Rubric for Showcase Evaluation:

Criteria	1 Point (Developing)	2 Points (Approaching)	3 Points (Proficient)	4 Points (Exemplary)
Circuit Assembly	Connections are loose or incorrect; circuit does not function.	Some correct connections, but needs significant help to function.	Circuit is assembled correctly and functions with minor guidance.	Circuit is assembled perfectly, functions reliably, and shows neat wiring.
Component Placement	Button and buzzer are placed randomly, not matching the path plan.	Components are placed generally near planned spots but not precisely.	Components are placed in the pre-selected spots from the planning activity.	Components are strategically placed in optimal spots, enhancing usability of the path.
Functionality	Buzzer does not make a sound when the button is pressed.	Buzzer makes a sound intermittently or weakly.	Buzzer consistently makes a sound when the button is pressed.	Buzzer consistently makes a clear, distinct sound with appropriate volume when pressed.
Understanding	Cannot explain how the circuit works or its purpose.	Can vaguely describe how the circuit works but struggles with purpose.	Can explain how the button and buzzer work together to create a sound cue.	Clearly articulates how the circuit functions and its importance for guiding people who cannot see.
Safety & Care	Requires frequent reminders for safe handling of components.	Handles components with some care but occasionally forgets.	Handles all circuit components and tools safely and responsibly.	Demonstrates exemplary safety practices and careful handling of all materials throughout the build.

It's time to move to the improve part and add this:

Now I Can

Objective: Students will reflect on how they applied their understanding of sound, safety, and math to support visually impaired individuals.

Activity Instructions:

- Introduce the Purpose
  - o Say: "Now that we've finished building our sound-guided path, let's take a moment to think about everything we learned and practiced."
  - o Explain that this activity helps students recognize what skills and knowledge they used.
- Read Each Statement Aloud
  - o Read each "Now I Can" statement slowly and clearly.
  - o Pause after each one to give students a moment to think.
- Student Response Options
  - o Have students check the box next to each statement they feel confident about.
- Optional Sharing
  - o Let students share one "Now I Can" statement with a partner or in a small group.
  - o You may ask: "Which one are you most proud of?" or "Which one was hard at first but easier later?"
- Wrap-Up Reflection
  - o Ask students to look at their checklist and reflect:

  "Which skill do you want to practice more?"

  "How do you feel helping others with your design?"

- Collect and Review
  - Use the checklists to informally assess student understanding and guide review or celebration activities.

Relation to Standards:

1. NGSS (Next Generation Science Standards)

- 1-PS4-1

  Standard: Plan and conduct investigations to provide evidence that vibrating materials can make sound and that sound can make materials vibrate.

  Relation: Though focused on sound, this standard connects to how students observe cause-and-effect between energy and motion while testing how their circuit affects movement.

- K-2-ETS1-1 (Engineering Design)

  Standard: Ask questions, make observations, and gather information about a situation people want to change to define a simple problem.

  In this lesson:

  Students explore a real-world design challenge (making movement accessible).

  They ask: “How can we make it move safely and easily for everyone?”

- K-2-ETS1-2

  Standard: Develop a simple sketch, drawing, or physical model to illustrate how the shape of an object helps it function.

  In this lesson:

  Students create models of circuits or moving devices.

  They identify how component placement affects function.

- K-2-ETS1-3

  Standard: Analyze data from tests of two objects designed to solve the same problem to compare strengths and weaknesses.

  In this lesson:

  Students compare different circuit arrangements.

  They analyze which design performs better and explain why.

2. ISTE Standards for Students

- Empowered Learner (1a, 1c)

  Students set learning goals, use digital or physical tools to test their understanding, and reflect on outcomes.

  In this lesson: Students test how their design functions and use evidence to explain results.

- Innovative Designer (4a, 4b, 4c)

  Students apply a design process to identify problems, test solutions, and reflect on results to improve.

  In this lesson: Students follow the design–test–improve cycle when building and refining their motorized system.